Specification

1. Technical Field

The invention falls within the technical fields of artificial intelligence, multi-agent collaboration, semantic retrieval (RAG), data security and distributed ledgers. Specifically, it relates to a system and method for local right confirmation and incremental synchronization of cognitive assets based on offline local slicing vectorized hash fingerprints, incremental synchronization via distributed right confirmation ledgers, and session-level temporary read-only sandbox degaussing erasure.

2. Background Art

In Retrieval-Augmented Generation (RAG) and multi-agent systems, knowledge bases are core assets that determine reasoning quality. However, traditional cloud-based vectorization and retrieval schemes carry severe risks to security and ownership:

1. Leakage of user private plaintext data to the cloud: Existing RAG systems require users to upload local plaintext knowledge bases to cloud servers for chunking and feature encoding. This directly exposes users' core cognitive assets to third-party cloud service providers, and large model engines may retain such assets free of charge for their own secondary fine-tuning training, infringing upon user data sovereignty.

2. Lack of asset right confirmation and high synchronization overhead: Conventional local database change tracking mechanisms only record version logs, lacking property vouchers that can be anchored to decentralized networks. In scenarios such as intelligent agent revenue sharing or asset authorization, lightweight incremental synchronization protocols are absent to implement low-cost ownership verification.

3. Plaintext retention vulnerabilities at the computing power layer: During interactive reasoning, matched upper and lower plaintext fragments are permanently stored by large model service gateways or temporary caches after being transmitted to external large models, lacking fine-grained lifecycle constraints and physical memory-level tamper-resistant cleaning means.

Therefore, designing a closed-loop data right confirmation and protection system featuring "assets stored locally only, fingerprints uploaded incrementally to the ledger, sandbox isolation during usage, and complete degaussing clearing after use" is an urgent technical challenge in the interdisciplinary field of network security and artificial intelligence.

3. Summary of the Invention

3.1 Technical Problems to Be Solved

To overcome the drawbacks of conventional schemes including unauthorized data crawling, plaintext residue at the computing power layer, and difficulties in tracing incremental asset right confirmation, the invention proposes a system and method for local right confirmation and synchronization of cognitive assets, which realizes incremental asset registration without exposing plaintext data and completes fine-grained physical destruction at the reasoning execution site.

3.2 Technical Solutions

To address the above technical problems, the invention provides a system for local right confirmation and incremental synchronization of cognitive assets based on hash fingerprints.

Aspect 1: A System for Local Right Confirmation and Incremental Synchronization of Cognitive Assets Based on Hash Fingerprints

The system comprises:
Local Security Zone, deployed on local physical equipment, configured to perform offline slicing vectorization and feature extraction, conduct cryptographic coupling calculation on feature vectors, plaintext contents and signature salt values by using a local private key to generate globally unique unidirectionally encrypted hash fingerprints, and store original cognitive asset bodies in ciphertext only within the local equipment;
Distributed Incremental Right Confirmation Ledger, connected to the Local Security Zone, configured to receive and record incremental hash fingerprints and timestamps periodically scanned and reported by the Local Security Zone as ownership evidence of the original cognitive asset bodies, wherein the Distributed Incremental Right Confirmation Ledger does not hold any original plaintexts or original vector indexes;
and a **Computing Power Isolation Operating Environment**, deployed in a hardware Trusted Execution Environment (TEE), connected to the Local Security Zone via a read-only data channel, provided with a session lifecycle counter and a physical degaussing erasure module. When an external session call occurs, the environment applies for a temporary physical security sandbox to decrypt and load Top-K semantically associated plaintext fragments. When a session terminates or the session lifecycle counter resets to zero, a hardware security microcontroller forcibly triggers a physical degaussing erasure interrupt to dynamically perform multi-pass random noise physical

overwriting erasure on the physical memory of the sandbox loaded with private knowledge slice data, including clearing processor caches and disrupting charge metastable states of DRAM media.

Preferably, a unidirectional encryption fingerprint calculation unit inside the Local Security Zone splices bytewise fixed-dimensional feature vectors v_emb generated by a local vector embedding model and corresponding sliced plaintext contents by using a signature salt value generated via a local asymmetric private key, and invokes a hash algorithm for cascading to generate a 256-bit SHA-256 hash value as the hash fingerprint.

Preferably, the Distributed Incremental Right Confirmation Ledger only receives incremental hash fingerprint registration through lightweight data packets containing `{ H_chunk, Timestamp, Owner_Signature }`, where H_chunk denotes the hash fingerprint and Owner_Signature denotes the private key signature of the local device; transmission of any plaintexts and vector features is completely blocked during synchronization.

Preferably, the temporary physical security sandbox is a secure encrypted memory segment allocated based on hardware TEE enclave technology; the secure encrypted memory segment is mounted with read-only attributes and deprived of all network and physical medium write permissions except for inference channels of designated heterogeneous large model engines.

Preferably, when performing overwriting erasure, the physical degaussing erasure module invokes a hardware-level secure erasure interface to repeatedly overwrite pseudo-random noise on physical memory pages corresponding to the temporary physical security sandbox and reset physical charges to defend against cold-start residual memory reading attacks.

Preferably, a vector retrieval engine deployed inside the Local Security Zone is forcibly bound to the local Loopback address 127.0.0.1; an access control policy of the vector retrieval engine is set to only receive requests from user sovereign processes via local Inter-Process Communication (IPC), physically blocking direct external network and cross-namespace semantic retrieval probing against the vector database.

Preferably, the system further comprises a **Kernel-State Network Interception Module**, configured to monitor network sockets of external session calls and forcibly send TCP RST packets to target communication terminals when a session terminates or the lifecycle counter resets to zero, so as to erase transmission traces in network caches.

Preferably, when slicing plaintext contents offline, the Local Security Zone takes the binary bit sequence of the hash fingerprint as an input seed of a pseudo-random generator to dynamically adjust synonym replacement probabilities of preset word classes or punctuation arrangement features in sliced texts, so as to embed tiny grammatical deviation watermarks recognizable via statistical matrix inversion, serving as parameter space watermark judgment evidence against illegal crawling and training by large models.

Aspect 2: A Method for Local Right Confirmation and Incremental Synchronization of Cognitive Assets Based on Hash Fingerprints

Executed by a computer processor, the method comprises the following steps:
Offline Local Fingerprint Generation: the Local Security Zone slices a local plaintext knowledge base into semantic blocks and extracts feature vectors in an offline state, couples the feature vectors, plaintext contents and signature salt values to generate globally unique unidirectionally encrypted hash fingerprints, and stores original cognitive asset bodies in local ciphertext;
Distributed Incremental Synchronization and Registration: an incremental scanning detection module reports newly generated hash fingerprints incrementally to the distributed right confirmation ledger for registration and evidence storage upon detection;
and **Session-Level Isolation and Degaussing Erasure**: when a session call occurs, a read-only security sandbox is created in a hardware Trusted Execution Environment to decrypt and load Top-K semantically associated plaintext blocks, and a lifecycle clock is activated simultaneously; when a session terminates or the lifecycle clock resets to zero, the physical degaussing erasure module is triggered to dynamically perform multi-pass random noise physical overwriting erasure on the physical memory of the sandbox loaded with private knowledge slice data, including clearing processor caches and disrupting charge metastable states of DRAM media.

Preferably, in the step of offline local fingerprint generation, the generation formula of the hash fingerprint is:

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H_chunk = SHA256(v_emb || TextContent || Salt)
```

where v_emb denotes a feature vector, TextContent denotes plaintext contents, and Salt denotes a signature salt value.

Aspect 3: A Local-End Operating Method in Local Right Confirmation and Synchronization of Cognitive Assets Based on Hash Fingerprints

Executed by a Local Security Zone device, the method comprises the following steps:

Offline Local Slicing and Vectorization: slice a local plaintext knowledge base into text segments in an offline state, invoke a local vector embedding model to generate feature vectors, and store bodies of the text segments in local ciphertext;

Local Cascaded Fingerprint Generation: splice feature vectors and plaintext contents of the text segments in a cascaded manner by using a signature salt value generated via a local private key, and calculate globally unique unidirectionally encrypted hash fingerprints;

and **Incremental Active Synchronization and Chain Upload**: periodically scan local right confirmation logs; upon detection of newly added hash fingerprints, assemble and actively send incremental reporting requests to the distributed right confirmation ledger. The reporting requests contain hash fingerprints, timestamps and property ownership signatures generated via the local private key, and transmission of plaintext contents and feature vectors is completely blocked during chain uploading.

Preferably, in the step of offline local slicing and vectorization, when slicing the plaintext knowledge base, the binary bit sequence of the hash fingerprint is taken as an input seed of a pseudo-random generator to dynamically adjust synonym replacement probabilities of preset word classes or punctuation arrangement features in sliced texts, so as to embed grammatical deviation watermarks recognizable via statistical matrix inversion.

Aspect 4: An Incremental Evidence Storage Method for a Right Confirmation Ledger in Local Right Confirmation and Synchronization of Cognitive Assets Based on Hash Fingerprints

Executed by physical nodes of a distributed ledger, the method comprises the following steps:

Reception of Incremental Reporting Packets: receive lightweight ownership reporting packets periodically sent by Local Security Zone devices, wherein the reporting packets only contain hash fingerprints, timestamps and ownership signatures;

Digital Signature Verification: decrypt and verify ownership affiliation of the hash fingerprints on the basis of the ownership signatures in the reporting packets by using public keys corresponding to the Local Security Zone devices;

and **Property Right Evidence Registration and Query**: upon successful verification, register and write the hash fingerprints and timestamps into a decentralized ledger database as unique

ownership indexes without storing or writing any original plaintexts and original vector indexes, and provide read-only ownership comparison verification for recorded fingerprints to an external data sovereignty policy central gateway.

Preferably, in the step of digital signature verification, time sequence comparison is performed on timestamps contained in received reporting packets to defend against replay attacks targeting property right evidence registration; property right evidence registration adopts a decentralized blockchain cascaded block structure for physical storage.

Aspect 5: A Computing Power Sandbox Isolation Execution Method in Local Right Confirmation and Synchronization of Cognitive Assets Based on Hash Fingerprints

Executed by a computing power isolation operating processor deployed in a hardware Trusted Execution Environment, the method comprises the following steps:
Allocation of Secure Read-Only Sandbox: upon reception of a session call request, apply to the Trusted Execution Environment for allocation of a temporary physical security sandbox, and establish a unidirectional read-only data transmission channel with a Local Security Zone device;
Decryption Loading and Clock Binding: acquire decrypted plaintexts of Top-K relevant text segments via the read-only data channel, load the plaintexts into secure encrypted memory segments corresponding to the physical security sandbox, and bind a hardware lifecycle timer to the secure encrypted memory segments;
and **Physical Degaussing and Network Circuit Break**: when session completion or timer expiration is detected, immediately perform multi-pass random noise physical overwriting erasure on the physical memory of the physical security sandbox to clear residual charges, including clearing processor caches and disrupting charge metastable states of DRAM media, and forcibly send TCP RST reset packets to the peer session terminal at the network protocol stack layer to erase data transmission traces.

Preferably, in the step of physical degaussing and network circuit break, the multi-pass random noise physical overwriting erasure comprises sequentially writing all-zero values, all-one values and random bit noise generated by a cryptographically secure pseudo-random number generator to all physical addresses occupied by the secure encrypted memory segments; after each pass of overwriting, a processor cache line flush instruction is invoked to force data in all levels of processor caches to be flushed back to physical memory pages for charge neutralization.

Aspect 6: A Computer-Readable Storage Medium

A computer-readable storage medium storing a computer program thereon, wherein the program, when executed by a processor, implements the method according to any one of Claims 9 to 16.

3.3 Beneficial Effects

The beneficial effects brought by the invention include:

1. Ultimate physical isolation of property rights (assets stored locally only): Original plaintext bodies and high-dimensional vector indexes are permanently stored within the Local Security Zone. Loopback and local IPC access control policies effectively block semantic retrieval hijacking and probing against vector databases from the network layer, realizing a genuine physical closed loop for data property rights;

2. Low-overhead incremental right confirmation: Only unidirectionally encrypted hash features and property signatures are transmitted for ledger upload, with all plaintexts and vector matrices fully shielded. This avoids network bandwidth bottlenecks and mathematically cuts off all paths for competitors to reverse-engineer plaintexts from the decentralized ledger;

3. Zero residue via physical-level degaussing: Hardware timer reset and boundary-triggered degaussing mechanisms are introduced. Multi-pass forced overwriting of physical DRAM pages with all-zero values, all-one values and pseudo-random bit noise, combined with clflush instructions for potential neutralization, physically eliminate cold-start side-channel charge extraction vulnerabilities and guarantee context security after reasoning;

4. Active evidence collection against illegal training: During local slicing, tiny grammatical deviation watermarks are covertly embedded using fingerprint seeds. If large model vendors illegally crawl user texts for secondary pre-training, their parameter spaces will inevitably fit such grammatical deviations, and right holders can obtain direct legal evidence via statistical matrix decoding;

5. Protection against circumvention via pure software distribution media: The independent claim for a computer-readable storage medium incorporates software packages, local right confirmation clients or firmware images encapsulating the present right confirmation and synchronization program into the direct protection scope of the patent, comprehensively strengthening rights enforcement effectiveness during commercial circulation.

4. Brief Description of the Drawings

Figure 1: Core physical topology diagram of the system according to the invention;

Figure 2: Flowchart of offline local slicing vectorization and unidirectional cascaded hash fingerprint generation;

Figure 3: Flowchart of session-level temporary read-only sandbox scheduling and physical degaussing erasure execution.

5. Detailed Description of Embodiments

5.1 Terminology Definition and Generalization of Sovereign Nouns

To ensure consistency of sovereign concepts in formal patent submission, the alignment relationship of core assets in the official patent document is defined as follows:

1. Data Sovereignty Policy Central Gateway: Represents an official sovereign security access statement and console portal, functionally corresponding to a standardized protocol parsing node;

2. Local Encrypted Cognitive Asset Library: Represents a local encrypted private knowledge repository and vectorization engine, functionally corresponding to a local secure computing environment;

3. Decentralized Right Confirmation Chain: Represents a distributed right confirmation ledger for recording incremental encrypted hash fingerprints, functionally corresponding to a distributed evidence storage circulation node.

As an ecological integration description of the local right confirmation and synchronization system, the system can be seamlessly deployed in a three-layer separated multi-agent system architecture consisting of a rule cognition layer, an isolated asset layer and a computing power execution layer (refer to the parent case patent related to the ecosystem "A System and Method for Operating Multi-Agent Systems Based on Three-Layer Separation"). The Local Security Zone of the present system physically corresponds to the isolated asset layer (i.e., physical sovereign storage area), while the Computing Power Isolation Operating Environment physically corresponds to a trusted isolated operating environment of the rule cognition layer (such as hardware-level TEEs, virtual machine-level confidential computing enclaves, etc.). Meanwhile, the hash fingerprints generated by the present solution can be tightly interconnected with the cascaded hash system of standardized cognitive protocol containers (refer to the ecosystem-related patent "A Human-Machine Collaboration System and Method Based on Standardized Cognitive Protocols"), realizing cross-layer right confirmation and evidence storage of cognitive rules, exclusive mantras and original cognitive asset bodies throughout the full lifecycle.

5.2 Offline Local Slicing Vectorization and Fingerprint Calculation

When a user stores a new data unit into the Local Encrypted Cognitive Asset Library, the processor performs local offline vectorization calculation without connecting to the public network.

1. The processor chunks input text to extract text segments. To prevent illegal crawling of data by large models as training sets, when slicing plaintext contents offline, the Local Security Zone takes the binary bit sequence of the generated hash fingerprint as an input seed of a pseudo-random generator to dynamically adjust synonym replacement probabilities of preset word classes (e.g., randomly replacing "collaboration" with "coordination") or punctuation arrangement features in sliced texts, so as to embed tiny grammatical deviation watermarks recognizable via statistical matrix inversion.
2. A local lightweight vector model is invoked to generate feature vectors v_emb.
3. v_emb, plaintext text and a Salt generated by a local asymmetric private key are concatenated in sequence and input into the SHA-256 algorithm to calculate a hash fingerprint.

The hash fingerprint serves as a digital right confirmation voucher in the invention. Meanwhile, to defend against hijacking probing across namespaces or virtual network devices locally, the listening address of the vector retrieval engine is forcibly locked to the Loopback address 127.0.0.1 with an access control list configured to only respond to sovereign process calls via local Inter-Process Communication (IPC) such as Unix Domain Sockets.

5.3 Incremental Ledger Registration and Multi-Gateway Verification

After detecting newly added hash fingerprints, the local process only uploads the hashes and their timestamps to the Decentralized Right Confirmation Chain. Meanwhile, under the access protocol specified by the Data Sovereignty Policy Central Gateway, the fingerprint can be exposed to external audit institutions as right confirmation evidence. The entire ledger upload and query verification process excludes all plaintext contents and feature vectors, completely blocking illegal snooping by cloud large model vendors.

After receiving lightweight ownership reporting packets periodically sent by Local Security Zone devices containing hash fingerprints, timestamps and ownership signatures, ledger physical nodes verify the signatures using corresponding public keys to confirm property ownership, and write the hash fingerprints and timestamps into a decentralized block-structured database. To defend against replay attacks, ledger nodes compare received timestamps with the current ledger consensus time sequence; requests with a time difference exceeding the threshold are judged as illegal replay requests and rejected.

5.4 Isolated Sandbox Degaussing Erasure and Cold-Start Defense

When an external large model call request arrives, the system retrieves local Top-K text segments and loads them into TEE physical sandbox memory pages, and activates a timer.
Upon session completion or timer expiration, a hardware security microcontroller immediately triggers secure_memzero_hardware physical degaussing erasure to clear memory.

Combined with the Kernel-State Network Interception Module sending TCP RST packets to forcibly cut off and reset transmission connections, no network data traces remain.

5.5 Vertical Industry Application Scenarios and Embodiments

The present right confirmation and synchronization system can be widely deployed in multiple industries with high privacy and property rights requirements in practical deployment. Typical deployment embodiments are listed below.
1. Industrial manufacturing scenario: In defect detection and process debugging of smart factories, industrial design drawings and core process parameters are stored in the Local Security Zone. The system executes offline local slicing vectorization and cascades to generate 256-bit hash fingerprints for incremental synchronization to a decentralized industrial control right confirmation ledger; when a cloud vision large model invokes reasoning, the read-only channel dynamically loads Top-K image defect feature fragments into an isolated sandbox, and immediately initiates 3 rounds of degaussing overwriting and cache flushing after reasoning completion or timeout to prevent leakage of process parameters. This scenario is deployed on a three-layer separated multi-agent system operating architecture (i.e., a rule cognition layer, an isolated asset layer and a computing power execution layer physically isolated from each other), and implemented based on the operating method according to any one of Claims 9 to 16 of the invention.
2. Government classified affairs scenario: In classified government document circulation and approval, top-secret documents are stored encrypted in the Local Security Zone, and hash fingerprints and timestamps are registered to a government compliance evidence storage right confirmation ledger via lightweight packets; when a government large model is invoked for content summarization or compliance review, decrypted loading is performed within a TEE isolated sandbox, and a hardware security microcontroller is forcibly triggered to initiate physical degaussing erasure interrupts upon lifecycle counter expiration or session termination to realize zero residue after use. This scenario is deployed on a three-layer separated multi-agent system operating architecture, and implemented based on the operating method according to any one of

Claims 9 to 16 of the invention.

3. Financial data scenario: In bank customer credit qualification risk control decision-making, the private asset bodies of customers are sliced offline in the Local Security Zone. Meanwhile, the binary bit sequence of the fingerprint is used as a seed to covertly embed subtle grammatical deviation watermarks into slices. Feature fingerprints are registered and written into a decentralized distributed right confirmation chain for ownership anchoring. During risk control reasoning, a read-only physical sandbox is allocated in the Computing Power Isolation Operating Environment to read-only load Top-K risk control fragments; upon session termination, the network interception module forcibly sends TCP RST packets to disconnect connections. If external large models illegally crawl such data for secondary training, right holders can obtain court right confirmation evidence by decoding watermark features. This scenario is deployed on a three-layer separated multi-agent system operating architecture, and implemented based on the operating method according to any one of Claims 9 to 16 of the invention.

5.6 Anti-Circumvention Feature Description

As a critical anti-circumvention design feature of the invention, failure to adopt the technical solution of "local fingerprint generation + incremental ledger upload + sandbox degaussing" described in the invention in practical deployment, and direct adoption of conventional cloud RAG or ordinary memory loading schemes without physical cleaning mechanisms will directly lead to the following technical defects and full breakdown of defense lines:

(1) Irreversible leakage of user private cognitive asset plaintext data during transmission and cloud vectorized storage, resulting in loss of data sovereignty control;

(2) Lack of lightweight cascaded hash vouchers during incremental updates of cognitive assets, making it impossible for the system to implement low-cost property right registration and tamper-proof generational tracing on decentralized ledgers;

(3) Physical memory residues at the computing power layer cannot be forcibly charge-neutralized, allowing attackers to recover resident sensitive knowledge fragments via cold-boot physical memory extraction attacks.

The invention tightly interlocks the above technical features to effectively block competitors from evading property right protection using conventional architectures, strengthening the technical necessity and inventiveness of the patent solution.

5.7 Equivalent Replacement Cover-All Paragraph

Finally, those skilled in the art may understand that core inventive concepts described in the invention such as "Local Security Zone", "Distributed Incremental Right Confirmation Ledger", "Computing Power Isolation Operating Environment" and "physical degaussing erasure" are not limited only to the specific software and hardware structural forms listed in the embodiments, but cover all equivalent replacements and logical variants foreseeable by those skilled in the art without departing from the essential spirit of the invention, all of which shall fall within the protection scope of the invention.